

Directions: Select **three** questions from Part 1 and **one** question from Part 2. For the chosen questions,

- Solve the problem to the best of your ability. Use your textbook, classmates, or any available internet resources to ensure you've found the correct answer.
- These concepts may not have been discussed in class and you should seek alternate resources to learn about the context for the problem. **Bolded terms** may help you find solution methods. Non-proof solutions generally require a solving program (*MATLAB, a matrix-capable graphing calculator*).
- Give a *brief* write up of how you solved the problem. Make sure to describe why you used any information that is not found directly in the textbook or class notes.
- You can (and should!) work with classmates and develop the same solutions as others in class. However, your write up must not be identical to anyone else's submitted work or no credit will be given.

Deadline for turning these in is 10/9/26 at 2:30 pm. Please turn in on paper in class or submit to my office mailbox.

1. Choose **three** of the following problems to complete.

A. From §1.8 of your text, you can select from #8, 12, 16, 18, 20, or 22

B. A rental car company has a fleet of 500 cars at three locations: Airport, North, and Valley. A car rented at one location may be returned to any other location, creating a **recurrence relation** that can be modeled by a **linear difference equation**. The fractions of cars returned to the three locations are recorded in the **migration matrix**

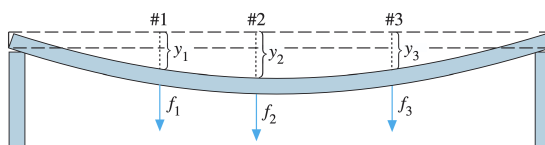
$$\begin{array}{ccc} \text{Airport} & \text{North} & \text{Valley} \\ \left[\begin{array}{ccc} 0.97 & 0.05 & 0.10 \\ 0.00 & 0.90 & 0.05 \\ 0.03 & 0.05 & 0.85 \end{array} \right] & \begin{array}{l} \text{Airport} \\ \text{North} \\ \text{Valley} \end{array} \end{array}$$

where the columns indicate which site the car was rented from and the rows indicate which site the car was returned to. For example, 97% of cars **rented from** the Airport site are **returned to** the Airport site. If on Monday there are 295 cars rented from the Airport site, 55 from the North location, and 150 from the Valley location, what will be the approximate distribution on Wednesday?

C. Let

$$D = \begin{bmatrix} 0.011 & 0.003 & 0.001 \\ 0.003 & 0.009 & 0.003 \\ 0.001 & 0.003 & 0.011 \end{bmatrix}$$

be a **flexibility matrix** for an elastic beam with flexibility measured in inches per pound.



- Suppose forces of 40, 50, and 30 *lbs* are applied at points 1, 2, and 3 respectively in the Figure. Find the corresponding deflections.
- Compute the **stiffness matrix** D^{-1} and list the forces needed to produce a deflection of 0.04 inches at point 3 with zero deflection at the other points.

2. Choose **one** problem from this section to solve.

- A. An economy has four sectors: Agriculture, Manufacturing, Services, and Transportation. Agriculture sells 20% of its output to Manufacturing, 30% to Services, 30% to Transportation, and retains the rest. Manufacturing sells 35% of its output to Agriculture, 35% to Services, 20% to Transportation, and retains the rest. Services sells 10% of its output to Agriculture, 20% to Manufacturing, 20% to Transportation, and retains the rest. Transportation sells 20% of its output to Agriculture, 30% to Manufacturing, 20% to Services, and retains the rest.
- Construct the **exchange table** for this economy using the concepts from the **Leontief input-output (or production) model**.
 - Find a set of **equilibrium prices** for the economy if the value of Transportation is \$10 per unit.
 - The Services sector launches a successful “eat farm fresh” campaign and increases its share of the output from the Agricultural sector to 40%, causing Manufacturing’s share of Agricultural production to fall to 10%. Construct the exchange table for this new economy.
 - Find a set of equilibrium points for this new economy if the value of Transportation is still \$10 per unit. What effect has the campaign had on the equilibrium prices for the different sectors in this economy?
- B. Consider some other useful matrix factorizations: *QR* factorization, Singular Value Decomposition, and Spectral Factorization. Prove two of the following:
- QR Factorization*: Suppose $A = QR$ where Q and R are $n \times n$, R is invertible and upper triangular, and Q has the property that $Q^T Q = I$. Show that for each $\vec{b}$ in $\mathbb{R}^n$, the equation $A\vec{x} = \vec{b}$ has a unique solution. What computations with Q and R produce the solution?
 - Singular Value Decomposition*: Suppose $A = UDV^T$ where U and V are $n \times n$ matrices with the property that $U^T U = I$ and $V^T V = I$, where D is a diagonal matrix with positive numbers $\sigma_1, \dots, \sigma_n$ on the diagonal. Show that A is invertible and find a formula for A^{-1} .
 - Spectral Factorization*: Suppose a 3×3 matrix A admits a factorization $A = PDP^{-1}$, where P is some invertible 3×3 matrix and D is the diagonal matrix

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Show that this factorization is useful when computing high powers of A . Find fairly simple formulas for A^2 , A^3 , and A^k (k a positive integer) using P and the entries in D .

- C. In a wind tunnel experiment, the force on a projectile due to air resistance was measured at different velocities,

Velocity (<i>100 ft/s</i>)	0	2	4	6	8	10
Force (<i>100 lbs</i>)	0	2.90	14.8	39.6	74.3	119

Use a solving program to find an **interpolating polynomial** - a polynomial whose path passes through every point - for these data that can be used to estimate the force on the projectile at unevaluated velocities. Use

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5.$$

What happens when you try to use a polynomial of degree less than 5? Predict the force when the projectile is traveling at *750 ft/s*.